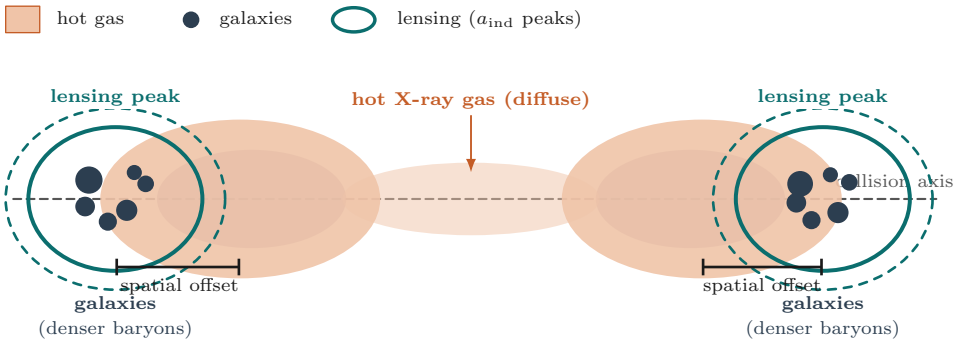


Bullet Cluster schematic: density-dependent a_{ind}



$$\Sigma_{\text{gal}}/\Sigma_{\text{gas}} \sim 3-5 \Rightarrow \kappa_{\text{ind,gal}}/\kappa_{\text{ind,gas}} \sim (3-5)^{0.4} \approx 1.55-1.90$$

Because a_{ind} scales with **local density**, lensing mass peaks with the galaxies (not the more diffuse post-collision gas).

No collisionless particle halo is required.